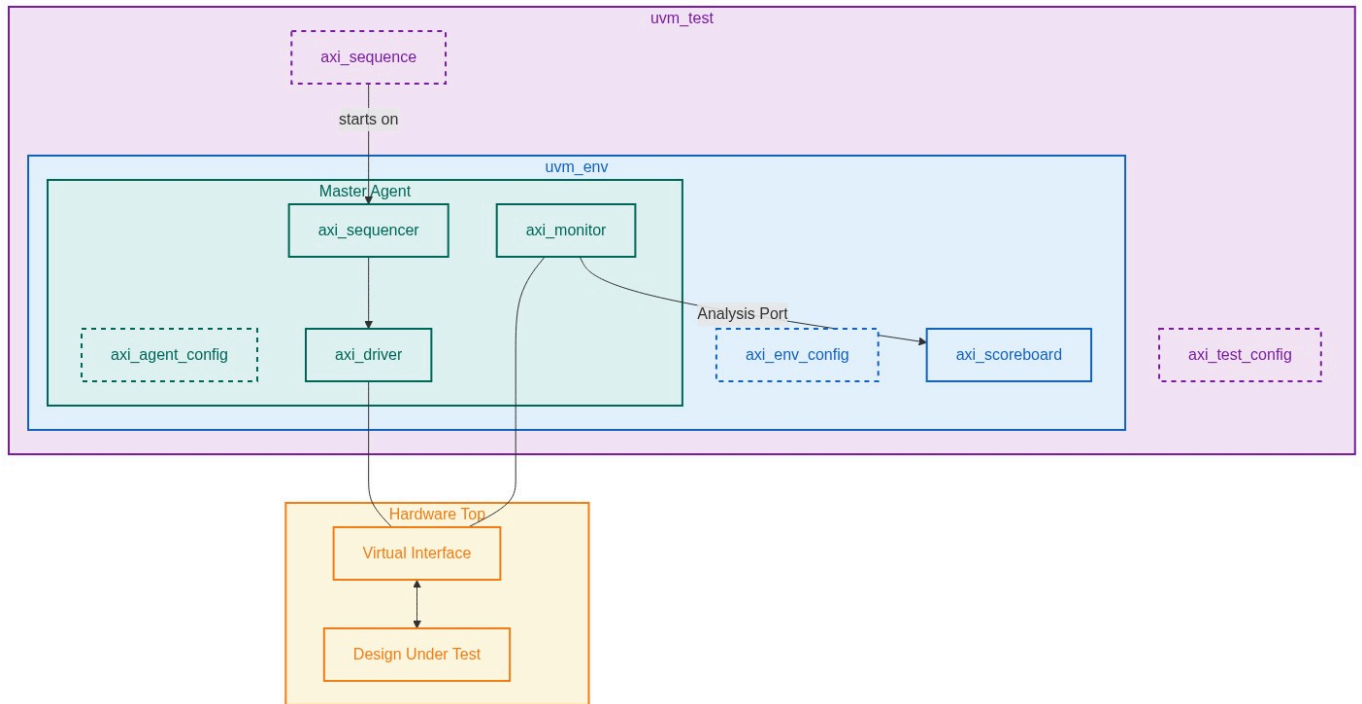


Verification Plan Report

Project: AXI-Stream Slave Verification Plan

Version: 3.2

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
Handshake & Stability	VALID/READY Flow Control	Impacts the driver's ability to hold T-data stable and the scoreboard's expectation of handshake cycles.
Interface Protection	AXI5 Parity (Odd)	Requires parity generation in the agent and checkers in the scoreboard to detect TDATACHK errors.
Data Streams	Sparse Streams (TKEEP/TSTRB)	Impacts the data packing logic in the scoreboard and the driver's sequence item complexity.
Packet Boundaries	TLAST Framing	Defines the end-of-packet conditions for the monitor's packet assembly logic.
Transfer Ordering	Continuous Packets	Affects the back-pressure scenarios and throughput performance measurements.

3. Test Requirements

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Index	Name	Description
REQ_HANDSHAKE_STABILITY	VALID and READY Handshake Stability	The Transmitter (Agent) must not wait for TREADY before asserting TVALID. Once TVALID is asserted, it must remain high until a handshake occurs. Verification must ensure the Slave DUT correctly samples data only on the rising edge of ACLK where both VALID/READY are high. Failure leads to lost or duplicated data.
REQ_AXI5_PARITY	AXI5 Interface Parity Protection	Odd parity checking is enforced for all signals. Each bit of the parity signal (TDATACHK) covers up to 8 bits of payload. Verification intent is to inject parity errors and confirm the DUT correctly identifies the fault, preventing corrupted data propagation.
REQ_PACKET_BOUNDARY	TLAST Framing Rules	TLAST must mark the final beat of a transfer. Verification must handle back-to-back packets and single-beat packets. Ensures the DUT correctly flushes or terminates processing at packet boundaries.
REQ_SPARSE_STREAM	TKEEP and TSTRB Handling	TKEEP (null-byte qualifier) and TSTRB (data-byte qualifier) define active bytes in each beat. Verification must handle sparse streams with "holes" to ensure the DUT reconstructs the data correctly.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_HANDSHAKE_SLAVE_WAIT	Slave (DUT) asserts TREADY first and waits for Master (Agent) to assert TVALID.	REQ_HANDSHAKE_STABILITY
TC_NULL_PACKET	Master sends a packet with TLAST=1 on the first beat and all TKEEP bits=0.	REQ_PACKET_BOUNDARY, REQ_SPARSE_STREAM
TC_AXI5_PARITY_ERROR_INJECTION	Agent injects a single-bit flip on TDATACHK during an active transfer.	REQ_AXI5_PARITY
TC_BACK_TO_BACK_PACKETS	Continuous stream where TLAST=1 is followed immediately by TVALID=1 for the next packet.	REQ_PACKET_BOUNDARY
TC_RESET_DURING_TRANSFER	Assert ARESETn while TVALID/TREADY are high; verify state machines return to IDLE.	REQ_HANDSHAKE_STABILITY

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5. Functional Coverage Groups

Group Name	Description
cg_handshake_scenarios	Cross of TVALID and TREADY transitions to cover all handshake timings.
cg_packet_size	Bins for single-beat, medium-burst, and maximum-burst packet sizes.
cg_sparse_keep_patterns	Walking 1s/0s and random toggle patterns on TKEEP/TSTRB signals.

6. Interface Signals

Signal Name	Direction	Width	Description
TVALID	Slave Input	1	Indicates that the Transmitter is driving valid data.
TREADY	Slave Output	1	Indicates that the Slave (DUT) can accept data.
TDATA	Slave Input	DATA_WIDTH	Primary data payload.
TLAST	Slave Input	1	Indicates the boundary of a packet.
TDATACHK	Slave Input	DATA_WIDTH/ 8	AXI5 Parity check bits for TDATA.

7. Verification Environment Structure

Root: uvm_axi_stream_slave_tb/

- |-- agent : Contains Master Agent (Sequencer, Driver, Monitor)
- |-- env : Contains Scoreboard and Environment configuration
- |-- tests : Contains UVM Test classes and sequence library
- |-- top : Hardware top-level with VIF and DUT